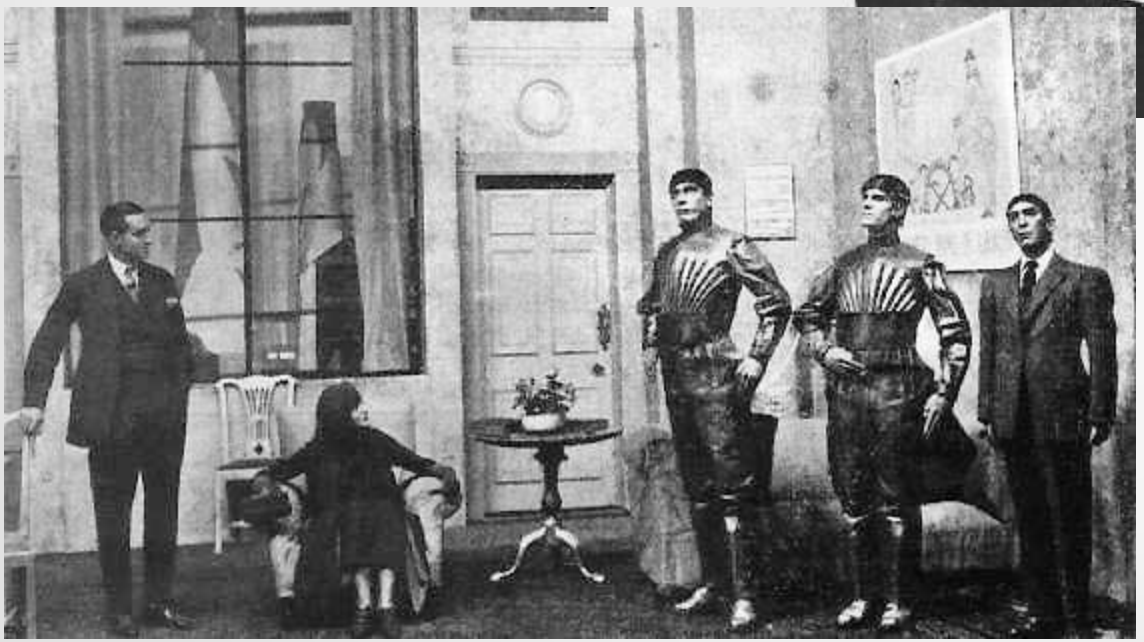
A brief
History of Rrobotics



Who came up with the word/[term?] ROBOT?

IN 1921,
CZECH WRITER
KAREL CAPEK
OFFICIALLY INVENTED THE TERM ROBOT TO DESCRIBE FICTIONAL AUTOMATA.



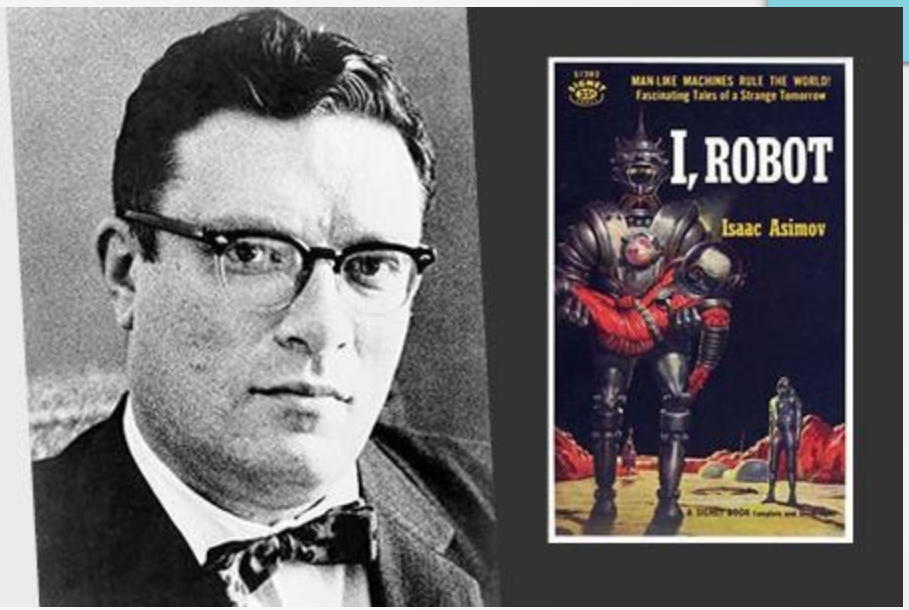


Karel Čapek 1890-1938 –Czech writer, prose writer and playwright in the 20th century, one of the pioneers of science fiction, who for the first time used and popularized the word robot which appeared in his R.U.R. (pl. Rossum's Universal Robots). The term robot, however, was actually coined by his brother Josef, who was a painter, poet, and author of children's books.

Robotics laws

1938 in his short story "Runaround" he used the word robotics for the first time.

IN 1942,
LEGENDARY WRITER
ISAAC ASIMOV
INVENTS THE TERM "ROBOTICS" AND FOREVER CHANGES SCIENCE FICTION WITH THE THE THREE LAWS OF ROBOTICS. THE FIRST LAW: "A ROBOT MAY NOT INJURE A HUMAN BEING OR, THROUGH INACTION, ALLOW A HUMAN BEING TO COME TO HARM."



Zero Law: A robot cannot harm humanity, nor can it, by neglect, cause harm to humanity.

First law: A robot cannot injure a human being, nor can it neglect a human being injured, unless it violates a law with a higher priority.

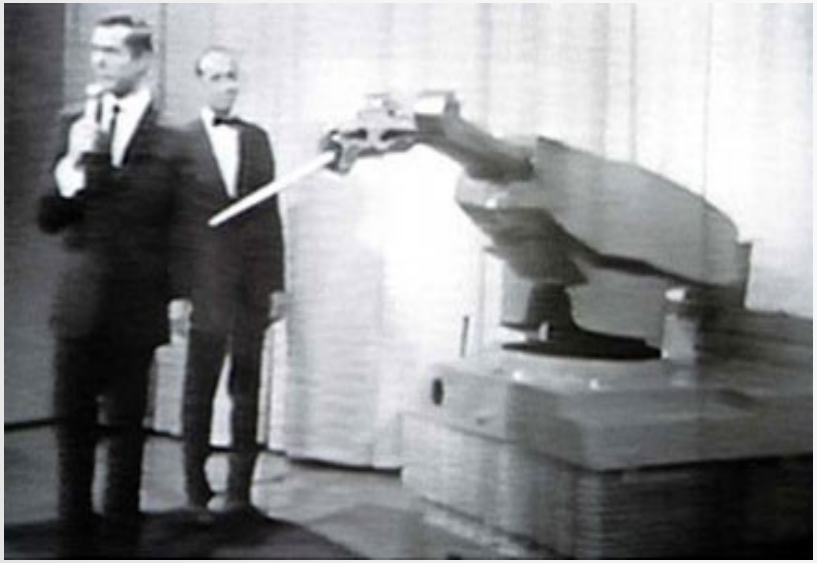
Second Law: The robot must obey human commands, except for commands that conflict with higher priority laws.

Third Law: The robot must protect itself as long as it does not conflict with the law of higher priority.

What is a ROBOT?

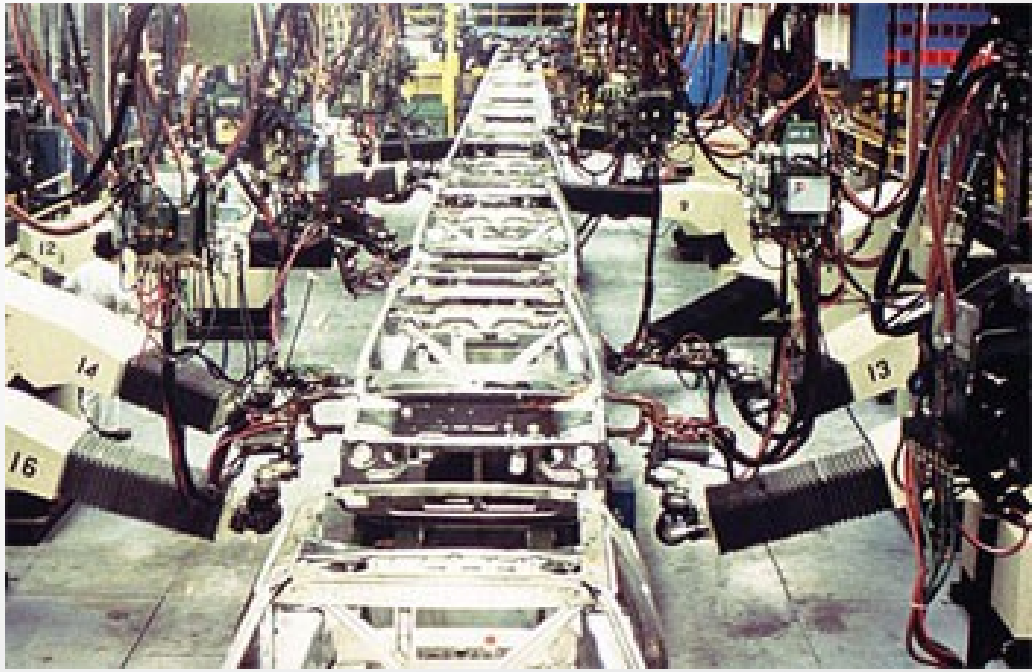
A robot is an automatically controlled, (re-) programmable, multi-task manipulation machine with many degrees of freedom, handling and locomotion properties, stationary or mobile, used for various industrial and special purposes.

1950 George Devol invented the Unimate, the first industrial robot. Unimate can transport die castings and weld them in cars. Like modern automation in manufacturing and other industries, these industrial robots would be programmed to a specific function as a means of replacing unskilled labor. Unimate was one of the most important milestones in the history of robots.



https://www.youtube.com/watch?v=zjPAgZ7Csjw&feature=youtu.be

... 20 years later:



And what will be the future? ...



Basis of robotics

- **Control theory**

A field dealing with the theory of mathematical analysis and modeling of objects and processes of various physical and social nature, treated as dynamically controlled systems. The model allows for the synthesis of the control system by introducing a controller that controls a given object or process so that it behaves in the desired manner. Other key concepts: feedback, dynamic correction/compensation.

- **Cybernetics**

Etymology: gr. κυβερνήτης kybernetes' steersman; steward ’from κυβερνᾶν kybernán, 'steer, control'; a transdisciplinary approach to control systems and related human-machine communication. The creator of cybernetics is the American mathematician Norbert Wiener.

- **Artificial intelligence**

https://en.wikipedia.org/wiki/William_Grey_Walter
https://en.wikipedia.org/wiki/Elmer_and_Elsie_(robots)

William Grey Walter (1910 – 1977) was an American-born British neurophysiologist, cybernetician and robotician.



Grey Walter's most well-known work was his construction of some of the first electronic autonomous robots. He wanted to prove that rich connections between a small number of brain cells could give rise to very complex behaviors - essentially that the secret of how the brain worked lay in how it was wired up. His first robots, which he used to call Machina speculatrix and named Elmer and Elsie, were constructed between 1948 and 1949 and were often described as tortoises due to their shape and slow rate of movement - and because they "taught us" about the secrets of organisation and life. The three-wheeled tortoise robots were capable of phototaxis, by which they could find their way to a recharging station when they ran low on battery power.

In one experiment, Walter placed a light on the "nose" of a tortoise and watched as the robot observed itself in a mirror. "It began flickering," he wrote. "Twittering, and jigging like a clumsy Narcissus."[citation needed] Walter argued that if it were seen in an animal it "might be accepted as evidence of some degree of self-awareness."

https://sites.google.com/view/machinaspeculatrix/home

One of the tortoises was modified, (given the pretend scientific name Machina docilis) and had added to its simple single celled "brain" one, then two conditional reflex circuits in which they could be taught simple behaviors similar to Ivan Pavlov's dogs.[citation needed] This tortoise was called CORA. One of these included being hit meant food whilst whistling means food, and when conditioned such a whistle by itself means being hit. When he added another circuit tuned to a whistle of another pitch, this could become whistle means being hit, whistle means food, this would make the animal become "afraid" whenever food was presented. Walter remedied this behaviour by severing the two additional circuits, and the tortoise reverted to being a Machina speculatrix. The conditioned reflex behaviour was later placed into a static desktop model, also known as CORA.

Elmer and Elsie (ELectroMEchanical Robot, Light-Sensitive) were two electronic robots that were built in the late 1940s by neurobiologist and cybernetician William Grey Walter. They were the first robots in history that were programmed to "think" the way biological brains do and meant to have free will. Elmer and Elsie were often labeled as tortoises because of how they were shaped and the manner in which they moved. They were capable of phototaxis which is the movement that occurs in response to light stimulus.

https://www.youtube.com/watch?v=6mysfrcOABY&list=RDLV6mysfrcOABY&index=1

The First Robot

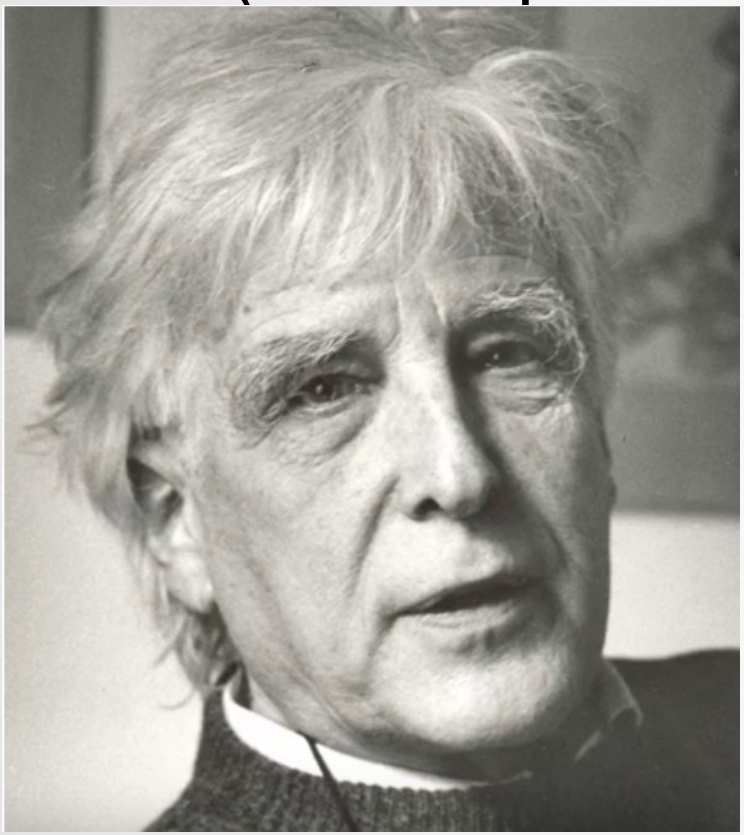
- William Grey Walter (1910 - 1977): neurophysiologist; interested in building biomimetic machines
- ***Tortoise*** robot:
  - tricycle-like design (one motor per wheel)
  - one photocell; one bump sensor
  - rechargeable battery
  - analog circuit and vacuum tubes connecting sensors with wheels



- Behaviors:
  - find light / head towards light / back away from light
  - turn and push to avoid obstacles
  - recharge battery
- Control: reactive, with prioritized reflexes
- First example of 'artificial life'; Tortoises now shown in museums around the world

Valentino Braitenberg

(next step in mobile autonomous robotics)



https://www.youtube.com/watch?v=k0algiyQW4Y
https://www.youtube.com/watch?v=DFXZc54_4GY
https://www.youtube.com/watch?v=A-fxij3zM7g

Valentino Braitenberg (1926-2011) Italian-Austrian cyberneticist. Models the animal world.
The Braitenberg 1980-84 vehicle is an agent that can move autonomously based on sensor data. It has primitive sensors that measure a stimulus at a point, and wheels (each driven by their own motor) that act as actuators or effectors.

In the simplest configuration, the sensor is directly connected to the effector, so that the detected signal immediately causes the wheel to move.

The first reactive robot (1984)

A Braitenberg vehicle is an agent that can autonomously move around based on its sensor inputs. It has primitive sensors that measure some stimulus at a point, and wheels (each driven by its own motor) that function as actuators or effectors. In the simplest configuration, a sensor is directly connected to an effector, so that a sensed signal immediately produces a movement of the wheel.

Depending on how sensors and wheels are connected, the vehicle exhibits different behaviors (which can be goal-oriented). This means that, depending on the sensor-motor wiring, it appears to strive to achieve certain situations and to avoid others, changing course when the situation changes.

The connections between sensors and actuators for the simplest vehicles (2 and 3) can be ipsilateral or contralateral, and excitatory or inhibitory, producing four combinations with different behaviours named fear, aggression, liking, and love. These correspond to biological positive and negative taxes present in many animals species.

Vehicle Getting Around

The vehicle has one sensor (e.g. a temperature detector) that directly stimulates its single wheel in a directly proportional way. The vehicle moves ideally in one dimension only and can stand still or move forward at varying speed depending on the sensed temperature. When forces like asymmetric friction come into play, the vehicle could deviate from its straight line motion in unpredictable ways akin to Brownian motion.

This behavior might be understood by a human observer as a creature that is 'alive' like an insect and 'restless', never stopping in its movement. The low velocity in regions of low temperature might be interpreted as a preference for cold areas.

Vehicle a

A slightly more complex agent has two (left and right) symmetric sensors (e.g. light detectors) each stimulating a wheel on the same side of the body. This vehicle represents a model of negative animal tropotaxis. It obeys the following rule:

More light right → right wheel turns faster → turns towards the left, away from the light.

This is more efficient as a behavior to escape from the light source, since the creature can move in different directions, and tends to orient towards the direction from which least light comes.

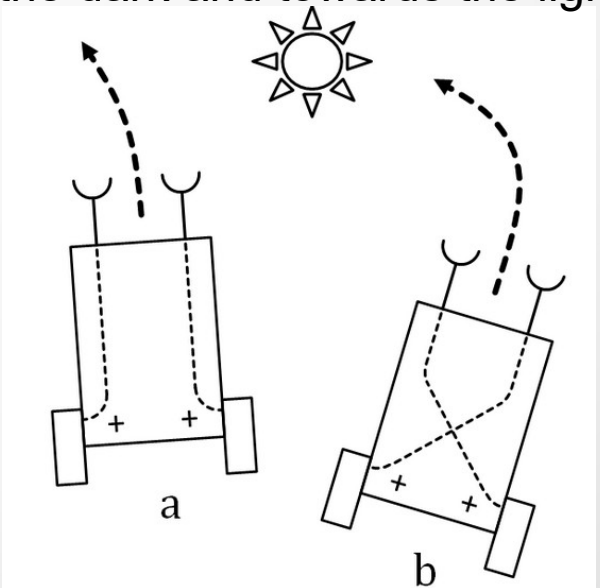
In another variation, the connections are negative or inhibitory: more light → slower movement. In this case, the agents move away from the dark and towards the light.

Vehicle b

The agent has the same two (left and right) symmetric sensors (e.g. light detectors), but each one stimulating a wheel on the other side of the body. It obeys the following rule:

More light left → right wheel turns faster → turns towards the left, closer to the light.

As a result, the robot follows the light; it moves to be closer to the light.



Artificial Intelligence

- Origin: in 1956 at a workshop at Dartmouth University, USA
- ***Dartmouth Summer Research Project on Artificial Intelligence***
- Some of the researchers present: Marvin Minsky, John McCarthy, Allan Newell, Herbert Simon, Claude Shannon, John Nash, etc…
- Outcome of meeting - what we need for AI is:
  - Internal models of the world
  - Search through possible solutions
  - Planning and reasoning to solve problems
  - Symbolic representation of information
  - Hierarchical system organization
  - Sequential program execution

Testing sensors and motors

1. Purpose of the research
2. Properties of research objects - description and characteristics {Wi}
3. Relationships between properties and external factors: {Wj} = f {Z}
4. Relationships between properties: {Wj} = f {Wm}}
5. Preparation of the conditions for the experiment
6. Testing => we collect the data
7. Presentation of the results.
8, Conclusions

EXAMPLES OF RELATIONSHIPS AND THEIR PRESENTATION

1. Distance measurements and actual measurements:

If $\{O_Z\}$ is a set of measurements, and $\{O_R\}$ is a set of real distances, what is the relationship:

$$O_Z = f(O_R)$$

? ie how much can you count on the sensor data?
Essentially, the measurement is expected to be true, but it is worth checking.

2. Can you count on the accuracy and repeatability of engine operation?
If $\{R_T\}$ is a set of tachometer measurements, and $\{R_I\}$ is a set of set revolutions, then what is the relationship

$$R_T = f(R_I)$$

? You can investigate the dependency of the difference
$D_{ti} = R_{Ti} - R_{ii}$ from power: